Course Syllabus for ME 264 – Introduction to Manufacturing for Mechanical Design

Course Information

ME 264 – Introduction to Manufacturing for Mechanical Design

- **Lecture:**
  Tuesdays and Thursdays LMBS 3239
  Section 1: 9:30 am – 10:20 am
  Section 2: 10:30 am – 11:20 am
  Section 3: 11:30 am – 12:20 pm

  Labs:
  Each section will publish a unique lab schedule. Labs will meet once a week on 3-hour blocks. All labs will meet in their assigned room. TAs/Lab Coordinators will provide initial instructions for the week, and then each subgroup will move to the corresponding Design Studio in the ME Building or Potter.

  Prerequisite: *CGT 163 -- Graphical Communication and Spatial Analysis or similar course*
- **Course credit hours: 3**
- Course Brightspace page: https://purdue.brightspace.com/d2l/home/1367675

Instructor(s) Contact Information

Instructors:

Dr. Mike Sealy

Dr. Martin Byung-Guk Jun
Office Hours: By appointment

Dr. Francisco Montalvo (Course Coordinator)

All lab sections will have a Lab Coordinator (Grad TA) and an undergraduate TA. For any questions about the lab activities, project and grading weekly lab reports, please get in touch with your Lab Coordinator.

Course Description

Introduction to manufacturing processes for typical engineering materials. Basics of material properties and behavior for manufacturing processes. Material selection and manufacturing process selection for mechanical design. Hands-on projects in the lab reinforce the knowledge of manufacturing processes.

Learning Resources, Technology & Texts

- Brightspace page
- Dedicated Laptops in POTR 333 (CAD/CAM software)
- Manufacturing Handbooks as needed (TBA)

You can access the course via Brightspace. It is strongly suggested that you explore and become familiar with the site navigation and the content and resources available for this course.

Learning Outcomes

Upon completion of this course, the student should be able to:

1. Understand material properties and behavior for design and manufacturing
    - Methods of evaluation: lab deliverables and project deliverables
2. Gain a fundamental knowledge of manufacturing processes.
    - Methods of evaluation: lab deliverables and project deliverables
3. Gain knowledge of the practice of measurements and manufacturing processes through laboratory experiments.
    - Methods of evaluation: lab deliverables and project deliverables
4. Enhance knowledge on the selection of materials and manufacturing processes for product design
    - Methods of evaluation: pre-labs, presentations, and project deliverables
5. Enhance technical communication skills through short technical lab reports.
    - Methods of evaluation: pre-labs, presentations, and project deliverables

Assignments

Your learning will be assessed through a combination of lecture and lab participation, lab deliverables, oral and written reports spread throughout the academic period. Details on these assignments, including a schedule of due dates and guidelines on discussion participation and evaluation, will be posted on the course website.

Assignments	Points
Lab Deliverables (10 total)	40 points
Project Reports and Presentations (Design Proposal and Final Report/Presentation)	40 points
Prelab Assignments	10 points
Quizzes	5 points
Team and Class Participation (Peer evaluation and Instructor Judgement)	5 points
NOTE: You must complete all labs and projects to pass this course. **Total**	**100 points**

Attendance Policy

This course follows the University Academic Regulations regarding class attendance, which state that students are expected to be present for every meeting of the classes in which they are enrolled. Attendance will be taken at the beginning of each class and lateness will be noted. When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, you should inform me of the situation as far in advance as possible. For unanticipated or emergency absences when advance notification to is not possible, contact me as soon as possible by Purdue email or phone. For absences that do not fall under excused absence regulations (see below), this course follows the following procedures:

1. Attendance to Lecture and Labs is mandatory. The lectures will contain important prelab and project information. Failure to attend will affect your ability to perform in the lab.
2. Do not come to class if you are feeling ill, but DO email your Lab Coordinator (Grad TA) with the subject line: ME 26400 Absence and copy your undergraduate TA. We do not need details about your symptoms. Just let me

know if you feel ill and cannot attend class. If it is an emergency, please follow the University regulations on medical care (see below).

3. Unless it falls under the University excused absence regulations (see below), any work due should be submitted on time via our course Brightspace/Gradescope to your TA.
4. If that day's class involves assessed work such as manufacturing a part, or a group presentation, you (and your group) will propose a plan to your TA and Instructor on if and how you can make up the work, following the assignment guidelines. This plan must be done before the next class/lab period, so email your TAs in advance when you know you will miss class.
5. The most important consideration in any absence is how it will affect your achievement of the assignment objectives and the course learning outcomes.

For cases that fall under **excused absence regulations**, you or your representative should contact or go to the Office of the Dean of Students (ODOS) website to complete appropriate forms for instructor notification. Under academic regulations, excused absences may be granted by ODOS for cases of grief/bereavement, military service, jury duty, parenting leave, or certain types of medical care. The processes are detailed at the ODOS website, please plan accordingly.

Course Schedule

Please refer to the course schedule provided in Brightspace.

Academic Integrity

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on our course Brightspace under University Policies and Statements.

See the University Policies and Statements section of Brightspace for guidance on Use of Copyrighted Materials. Effective learning environments provide opportunities for students to reflect, explore new ideas, post opinions openly, and have the freedom to change those opinions over time. Students and instructors are the authors of the works they create in the learning environment. As authors, they own the copyright in their works subject only to the university's right to use those works for educational purposes Students may not copy, reproduce, or post to any other outlet (e.g., YouTube, Facebook, or other open media sources or websites) any work in which they are not the sole or joint author or have not obtained the permission of the author(s).

Nondiscrimination Statement

A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in our course Brightspace under University Policies and Statements.

Accessibility

Purdue University is committed to making learning experiences accessible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS)**.** TAO is available to all students at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and

other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

Emergency Preparedness

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

A link to Purdue's Information on Emergency Preparation and Planning is located on our Brightspace under "University Policies and Statements." This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System. I encourage you to download and review the *Emergency Preparedness for Classrooms document*.

The first day of class, I will review the **Emergency Preparedness plan for our specific classroom**, following Purdue's required Emergency Preparedness Briefing. Please make note of items like:

- The location to where we will proceed after evacuating the building if we hear a fire alarm.
- The location of our Shelter in Place in the event of a tornado warning.
- The location of our Shelter in Place in the event of an active threat such as a shooting.